Chair of Biomedical Physics
TUM School of Natural Sciences
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Technical University of Munich

Symposium on Artificial Intelligence in Medical Imaging (SAIMI) 2026, Bern, Switzerland

A Physics-guided Implicit Neural Representation for Streak Reduction in X-ray Dark-field CT

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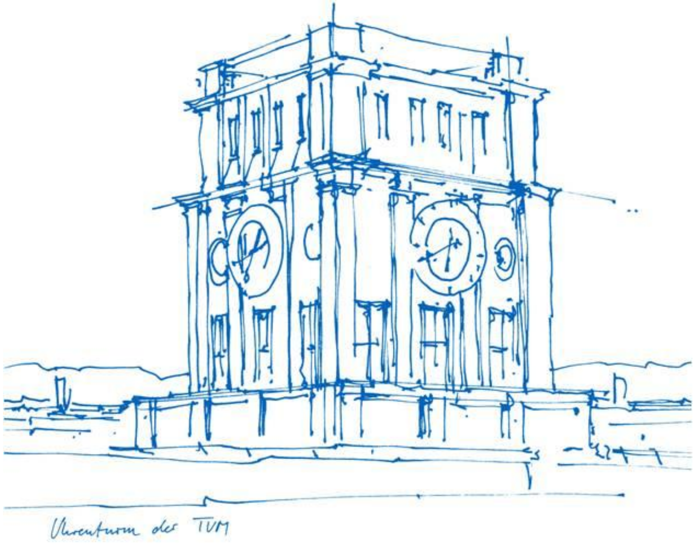
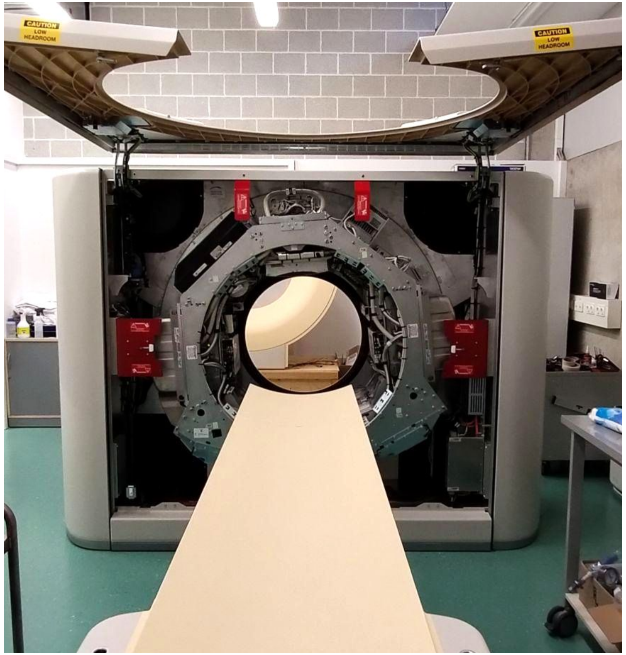
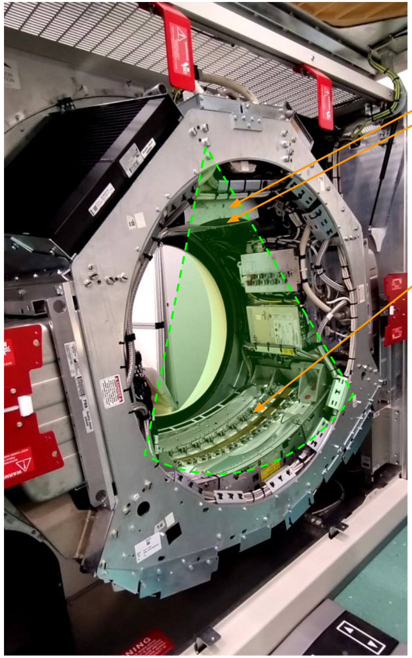
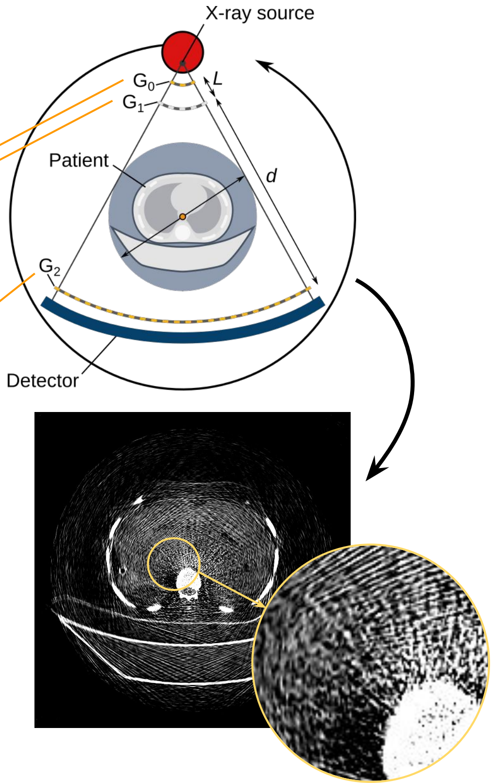


Background: Human-scale Dark-field CT Prototype

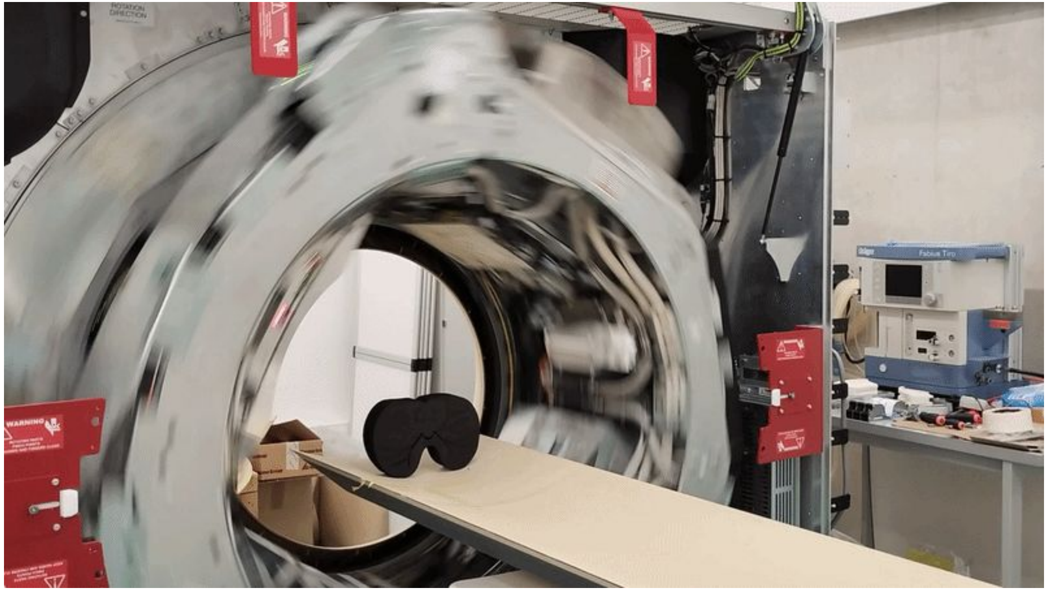






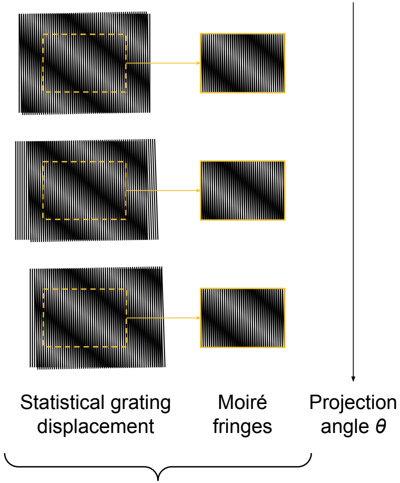
Philips iCT gantry operated at 550 mA, 1.5 s (825 mAs), 80 kVp

The Physics: Talbot–Lau Interferometry

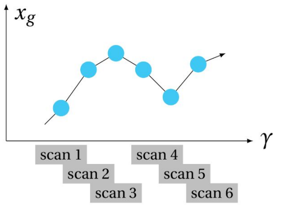


Air

Sample



Continuous acquisition:
Sliding-window phase retrieval

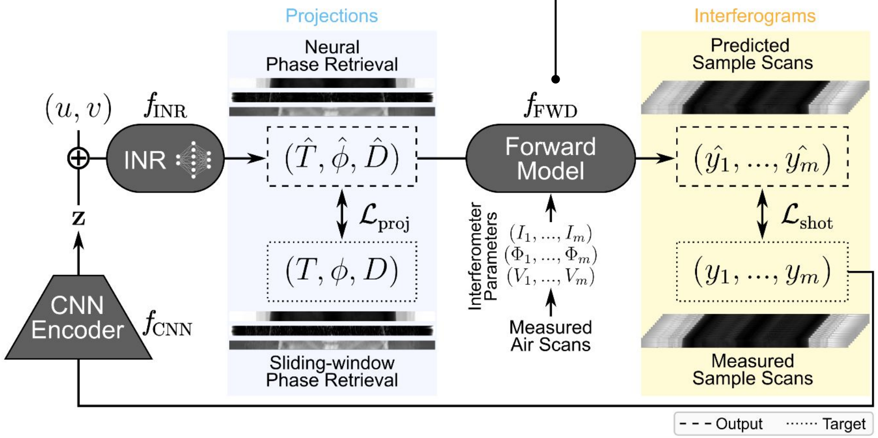


The Representation: Continuous Neural Field

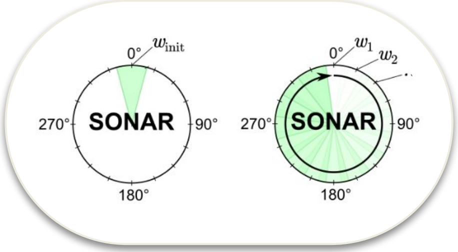
SONAR “Shot-Optimized Neural Adaptive Representation”

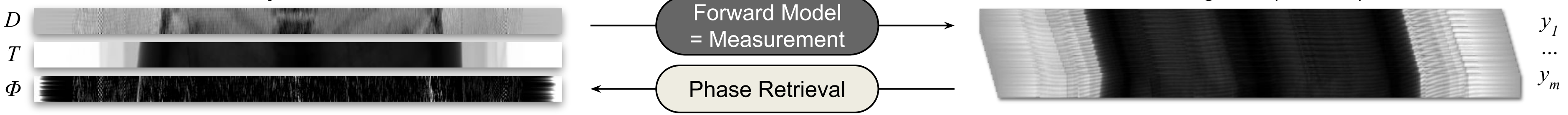
$$(T, \phi, D) \mapsto y = TI + TIDV\cos(\Phi + \phi)$$

Differentiable Talbot–Lau interferometer forward model:
Truncated Fourier series

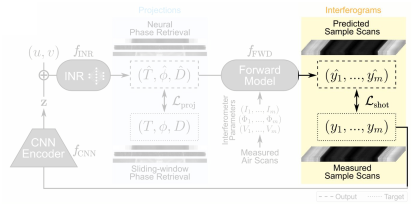


Training: Warm-start fine-tuning
Initialization shot: 1000 iterations
Shot-wise fine-tuning: 50 iterations





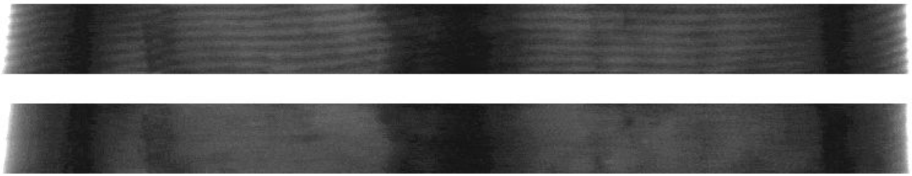
Measurement Space



Shots (2400 across the full CT rotation)

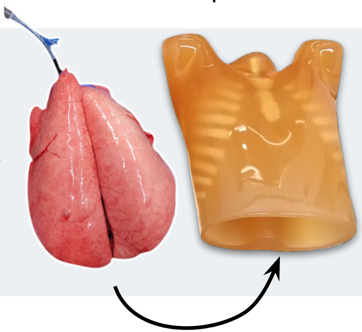
Conventional

SONAR



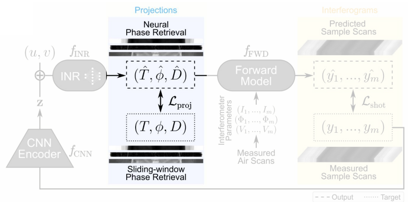
Measured sample:

Ex vivo porcine lungs Ventilated at 10 to 30 mbar

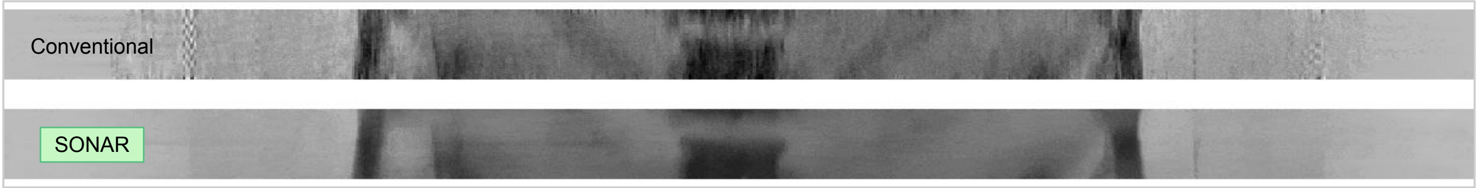


Human thorax phantom “Lungman”

Projection Space

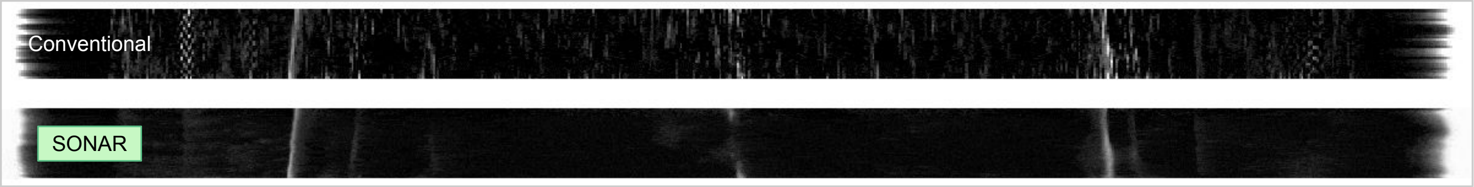


Dark-field

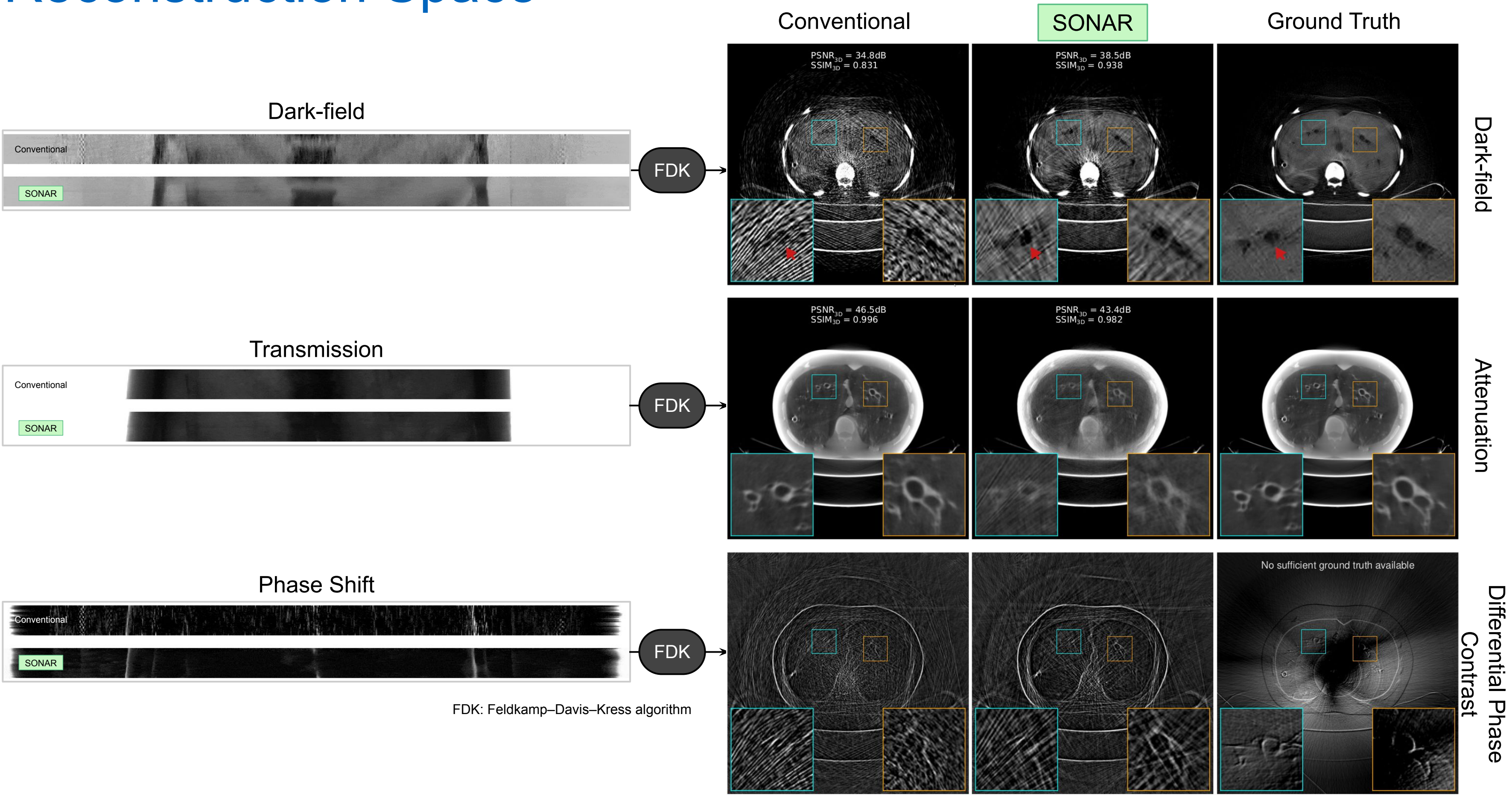


Transmission

Phase Shift

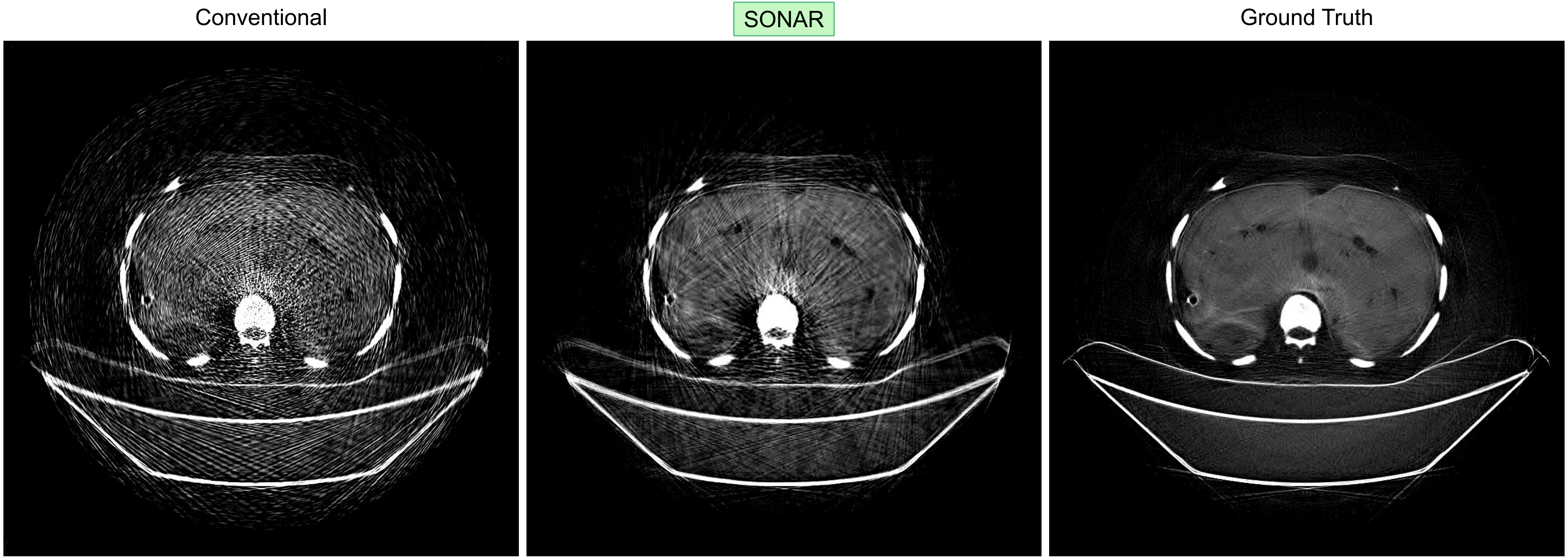


Reconstruction Space



FDK: Feldkamp–Davis–Kress algorithm

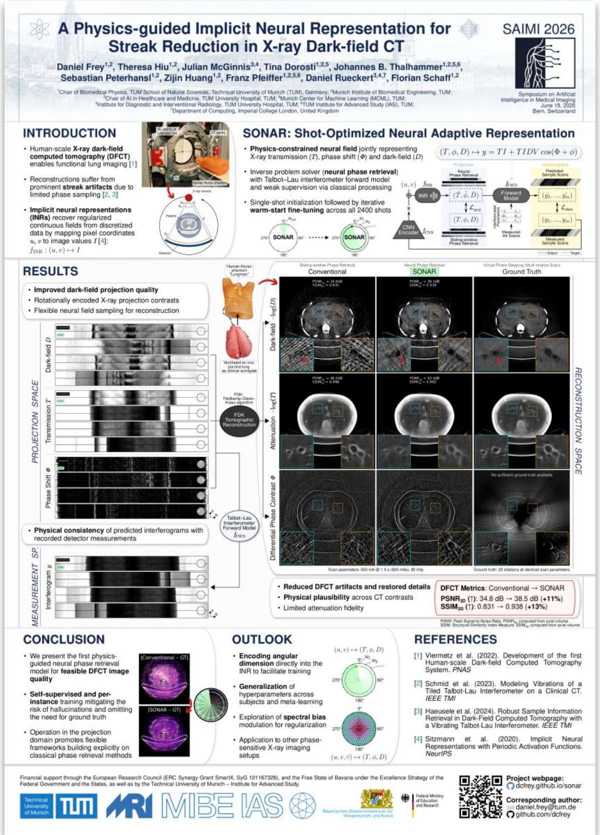
SONAR-enhanced DFCT

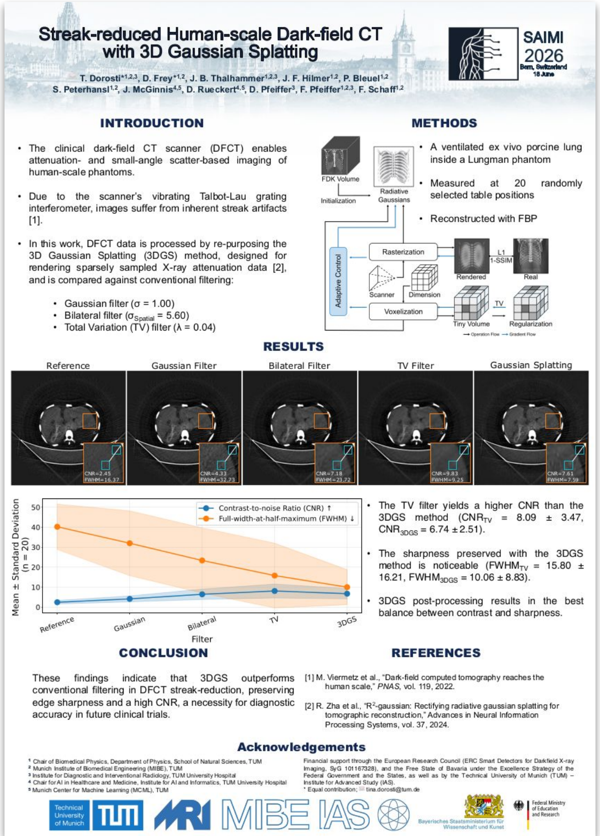


DFCT Metrics: Conventional → SONAR

$PSNR_{3D}$ (↑): 34.8 dB → 38.5 dB (**+11%**)
$SSIM_{3D}$ (↑): 0.831 → 0.938 (**+13%**)

More DFCT? Visit Our Poster Sessions





Thank you!

Project webpage:
dcfrey.github.io/sonar

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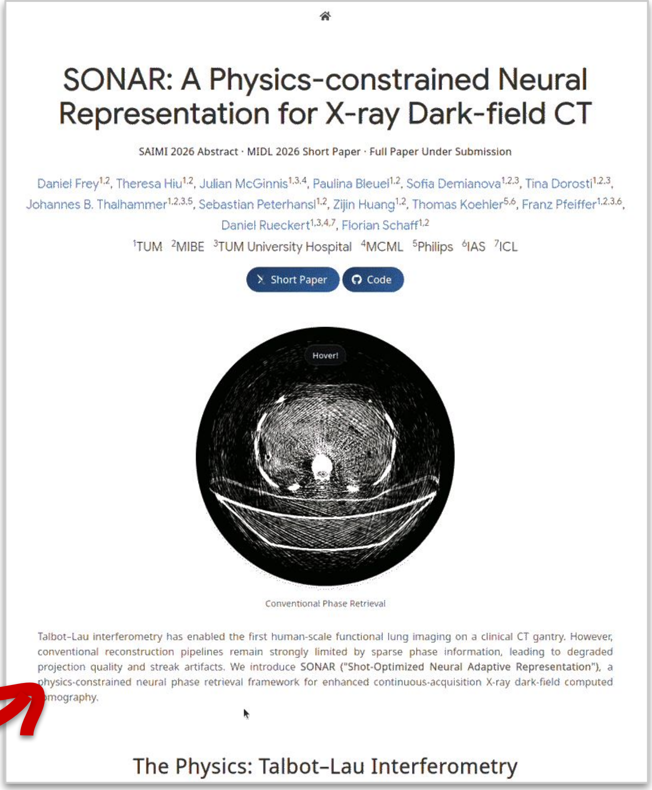

SONAR: A Physics-constrained Neural Representation for X-ray Dark-field CT

SAIMI 2026 Abstract · MIDL 2026 Short Paper · Full Paper Under Submission

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Short Paper   Code

Hover!

Conventional Phase Retrieval

Talbot-Lau interferometry has enabled the first human-scale functional lung imaging on a clinical CT gantry. However, conventional reconstruction pipelines remain strongly limited by sparse phase information, leading to degraded projection quality and streak artifacts. We introduce SONAR ("Shot-Optimized Neural Adaptive Representation"), a physics-constrained neural phase retrieval framework for enhanced continuous-acquisition X-ray dark-field computed tomography.

The Physics: Talbot-Lau Interferometry

PHILIPS Healthcare

Technical University of Munich

Bayerisches Staatsministerium für Wissenschaft und Kunst

Federal Ministry of Education and Research